HR Analytics Dashboard — Complete Rebuilding Guide

Power BI and Excel/CSV | Portfolio documentation

1. Prepare the project

1. Install Power BI Desktop and Microsoft Excel. Keep HR_Analytics-DataSet.csv in a stable local project folder.

2. Create folders: data/, powerbi/, docs/, images/. Keep the original file unchanged and work on a copy.

3. Open the CSV in Excel only for inspection; do not convert numeric IDs or categories to dates.

2. Import the CSV in Power BI

1. Home > Get data > Text/CSV > select HR_Analytics-DataSet.csv > Transform Data.

2. In Power Query rename the query HR_Analytics. Set EmpID, AgeGroup, Attrition, Department, JobRole, SalarySlab, Gender, EducationField to Text. Set numeric fields to Whole Number except YearsWithCurrManager to Decimal Number.

3. Use View > Column quality and Column profile to inspect missing and error values. Keep the original 1,480-row query as HR_Raw and make a Reference query called HR_Analytics for the dashboard.

3. Resolve source quality issues

1. Sort EmpID ascending, then Home > Remove Rows > Remove Duplicates on EmpID in the dashboard query. This retains one record per employee. Confirm whether duplicate IDs have conflicting details before choosing which record to retain.

2. Normalize BusinessTravel: replace TravelRarely with Travel_Rarely. Do not change other category labels without evidence.

3. Leave null YearsWithCurrManager as null. Record that 57 original rows lack this value.

4. Close & Apply. In Data view confirm 1,470 rows and the expected categories.

4. Create calculated columns

1. In Data view choose New column. Enter the two DAX columns below only if you need explicit row-level attrition counts.

2. The source already contains AgeGroup and SalarySlab, so use those fields rather than inventing new group boundaries.

5. Create measures and validate

1. In Modeling > New measure create the measures below. Put each in a separate measure.

2. Create a temporary table visual showing each measure and compare to the verified baseline.

3. Set Attrition Rate format to Percentage with two decimal places, Avg Age and Avg Tenure to two decimals, Avg Monthly Income to two decimals.

6. Create the six KPI cards

1. Insert > Card. Add Headcount; duplicate the visual for Attrition Rate, Avg Age, Attrition Count, Avg Monthly Income and Avg Tenure.

2. Use the Format pane to set display units, titles and data labels. Match the PBIX report page arrangement.

3. For headcount, use distinct EmpID, not raw row count or SUM(EmployeeCount).

7. Rebuild the visuals

1. Donut chart: Legend EducationField; Values Attrition Count (or Headcount for workforce distribution, clearly label which).

2. Clustered column chart: X-axis AgeGroup; Y-axis Attrition Count; sort age groups logically (18–25, 26–35, 36–45, 46–55, 55+).

3. Matrix: Rows JobRole (or another source-backed grouping); Columns JobSatisfaction; Values Attrition Count. The PBIX confirms JobSatisfaction as columns; inspect the original visual for exact rows.

4. Clustered bar: Y-axis SalarySlab; X-axis Attrition Count. Set SalarySlab sort order manually.

5. Clustered bar: Y-axis JobRole; X-axis Attrition Count; sort descending.

6. Area chart: X-axis YearsAtCompany; Y-axis Attrition Count; use numeric continuous axis where appropriate.

7. Treemap: Group Gender; Values Attrition Count. Use descriptive titles so readers know whether charts show attrition or headcount.

8. Create and connect slicers

1. Insert > Slicer. Drag Department into the Field well.

2. In Format > Slicer settings select List or Tile, then enable single/multiple selection as intended.

3. Select slicer > Format ribbon > Edit interactions. For every chart/card choose Filter or None; choose Filter for all workforce metrics.

4. Test each department (Research & Development, Sales, Human Resources). Confirm every intended visual responds.

5. For multi-page reports, use View > Sync slicers and select pages; this supplied PBIX has only one report page.

9. Page layout and design

1. View > Page view > Fit to page. Use Format canvas settings to choose a consistent 16:9 page size.

2. Use aligned KPI cards across the top and supporting visuals below; apply a limited palette and readable font sizes.

3. Use the original PBIX theme if available; do not claim pixel-perfect reproduction from metadata alone.

10. Refresh and full-screen

1. Power BI Desktop: Home > Refresh updates imported data from the configured CSV path.

2. Power BI Service: after publishing, configure credentials and a supported gateway if the CSV remains local.

3. Power BI Service: use report Full screen from the report menu. Desktop offers Focus mode for visuals and Fit to page; no VBA is executed by PBIX.

4. Optional Power BI button: Insert > Buttons > Blank; set Action to Bookmark or Page navigation. A bookmark can control visibility/navigation, but cannot run VBA or trigger dataset refresh.

11. Optional Excel companion VBA

1. VBA works only in a macro-enabled Excel workbook (.xlsm), not inside a PBIX.

2. Excel: Developer > Visual Basic > Insert > Module, paste the example macros below.

3. Developer > Insert > Button (Form Control) > assign ToggleExcelFullscreen or RefreshExcelData.

4. Save as .xlsm, enable macros only when you trust the file, and test both buttons.

12. QA and publication checklist

1. Verify the distinct headcount of 1,470, attrition count 237 and attrition rate 16.12% after deduplication.

2. Confirm slicer selections change all intended visuals and reset to All.

3. Check blank values, sort orders, title/measure alignment and rounding.

4. Remove or replace employee-level identifiers before sharing source data publicly; check license and privacy requirements.

5. Export dashboard screenshot, add README.md, PDF documentation, guide and optionally a sanitized dataset to GitHub.

Exact reproducible DAX examples

Attrition flag

```
AttritionCount = IF(HR_Analytics[Attrition] = "Yes", 1, 0)
```

Headcount

```
Headcount = DISTINCTCOUNT(HR_Analytics[EmpID])
```

Attrition count

```
Attrition Count = CALCULATE(DISTINCTCOUNT(HR_Analytics[EmpID]), HR_Analytics[Attrition] = "Yes")
```

Attrition rate

```
Attrition Rate = DIVIDE([Attrition Count], [Headcount], 0)
```

Average age

```
Avg Age = AVERAGE(HR_Analytics[Age])
```

Average monthly income

```
Avg Monthly Income = AVERAGE(HR_Analytics[MonthlyIncome])
```

Average tenure

```
Avg Tenure = AVERAGE(HR_Analytics[YearsAtCompany])
```

Excel VBA examples (not recovered from the PBIX)

Fullscreen toggle

```
Sub ToggleExcelFullscreen()
    Application.DisplayFullScreen = Not Application.DisplayFullScreen
End Sub
```

Refresh all Excel data

```
Sub RefreshExcelData()
    ThisWorkbook.RefreshAll
End Sub
```

Calculation example

Using distinct employee IDs: 237 employees marked Yes ÷ 1,470 distinct employees = 0.161224... = 16.12%. If using all 1,480 rows, the raw proportion is 238 / 1,480 = 16.08%; this illustrates why the duplicate-handling rule must be documented.

Reconstruction boundary

The PBIX report layout and visual field references were inspected. The source model is stored in a compressed/binary DataModel; exact original DAX and transformation steps were not recovered. All DAX and VBA shown in this guide are reproducible implementation examples, not asserted copies of original code.